

Experimental Setup

Does the frozen radial Test 6 clock pass the longevity prerequisite before Test 7?

Experiment: gross.clock-longevity.v1 Model: signal-space.ss-ocf-1.flat-clock.v1

Runtime configuration: fixtures/research/gross-test-06-longevity.json

Plan lock: 210c2b2b5f6e9fcc0b749cb20cb2d2d3cb2543bf090b5e2445ba22fcdf72a854

MODEL AND NUMERICAL SETUP

Action: SS OCF 1 flat action, $U=s-s^2+s^3$, $V=0.25-0.4s+0.2s^2$, $\zeta=0.1$. Neutral response extension $Z=1+0.2s$ ($M0=m=1$); same underlying action, separate registered active-neutral sector.

Units: $c=\hbar=m=1$; r,t in m^{-1} ; fields in m ; frequencies in m ; dimensionless tick shift

$\Omega\delta t/(2\pi)$.

Domain: Saved $\omega_Q=0.900$ profiles; $R=80, h=0.1$ and $h=0.05$. Response $R=160$ independently. Fixed readout $r=0.1$.

Boundaries: Regular origin and zero radial fields at finite R . Longevity uses original logged sponge

beyond $r=60$. Response uses no sponge; earliest outer reflection cannot reach core by $t=100$.

Initial Data: Exact saved discrete profiles/eigenmodes; $\chi=0.001 f/\max(f/r)$, χ momentum=0, Φ momentum= $-i*0.9*\Phi$. Response $ra=A*c*\cos^4(\pi*(r-c)/(2*w))$ inside $|r-c|<w$, zero outside; $(c,w)=(18,3)$ incoming, optional (4,2) outgoing. Initial $d_t(ra)=+d_r(ra)$ incoming and negative outgoing;

$p_a=Z*d_t(ra)$. Amplitude A is the coefficient of this prescribed radial packet; it is not an exact peak a normalization.

Solver: Longevity reuses registered kick-drift-kick at $dt=.04,.02$; all 100 nominal periods, sample every

.2. Response uses RK4 of discrete reciprocal Hamiltonian at $dt=.02$, mesh/time .01; samples .1; $t=100$.

First solve exact Taylor coefficients $a_1, \delta\Phi_2, \delta\chi_2$ on evolving no-pulse baseline; save all predictions before full nonlinear runs. No parameter fit. Neutral stiffness $G^T Z_{\text{faces}} G$, G

$b=r_{\text{face}}*d_r(b/r)$.

Tolerances: Locked criteria below. L2 means Euclidean norm of uniformly sampled local traces over

$t=0..100$. Numerical error is direct coarse/fine difference, no assumed Richardson order. Deterministic error floor 0.2% for response approximation; unresolved if signal RMS $\leq 1e-12$. Resource ceiling 1800 s wall, 1024 MiB, 100 MiB output.

COMPETING PREDICTIONS

usable: The frozen calibration survives full refined longevity and predicts an even, amplitude-squared local response within registered error controls.

inadequate: Unstable clock, sign-odd response, persistent prediction residual or ambiguous local ticks invalidates the tested approximation.

CONTROLS

frozen: exclude branch or calibration retuning; method: verify saved profile/mode hashes; same physical

local radius $r=0.1$ on every grid

numerics: distinguish physical response from discretization and boundaries; method: independent $h/2$ at

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R=80, dt/2 at h=0.1; response additionally R=160; no change to input physical profile branch

PREREGISTERED DECISIONS

frozen-inputs (frozen-inputs)

Pass: All input SHA-256 values match the frozen Test 6 profiles and modes.

Fail: Finite, resolved data violates the preregistered inequality.

Unresolved: Missing, nonfinite or under-resolved signal/marker or required control.

Uncertainty: Direct independent mesh, time, box differences; finite sampling; no statistical confidence claim.

local-record (local-record)

Pass: Each 100-period trace has at least 99 positive-going crossings and frequency within 2% of frozen Ω .

Fail: Finite, resolved data violates the preregistered inequality.

Unresolved: Missing, nonfinite or under-resolved signal/marker or required control.

Uncertainty: Direct independent mesh, time, box differences; finite sampling; no statistical confidence claim.

lifetime (lifetime)

Pass: Absolute final projected mode-energy change <1% on base, h/2 and dt/2 over 100 nominal periods.

Fail: Finite, resolved data violates the preregistered inequality.

Unresolved: Missing, nonfinite or under-resolved signal/marker or required control.

Uncertainty: Direct independent mesh, time, box differences; finite sampling; no statistical confidence claim.

convergence (convergence)

Pass: Local frequencies differ by <1%; absolute final mode-energy fractions differ by <0.01 on mesh/time controls.

Fail: Finite, resolved data violates the preregistered inequality.

Unresolved: Missing, nonfinite or under-resolved signal/marker or required control.

Uncertainty: Direct independent mesh, time, box differences; finite sampling; no statistical confidence claim.

charge-ledger (charge-ledger)

Pass: Maximum charge-plus-sink relative residual <1% on every 100-period trace.

Fail: Finite, resolved data violates the preregistered inequality.

Unresolved: Missing, nonfinite or under-resolved signal/marker or required control.

Uncertainty: Direct independent mesh, time, box differences; finite sampling; no statistical confidence claim.

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VISUAL QUESTIONS

local-ticks [evidentiary]: Does the same local radius keep ticking over 100 periods on each grid?

View: PNG/SVG/PDF and exact CSV; observables: local clock, mode energy, control residuals

usable: matched prediction, parity, scaling and persistent ticks

inadequate: resolved failure or missing ticks

Controls: frozen, numerics; uncertainty: Independent numerical controls; spherical restriction

Data: plot-data/summary.json, plot-data/traces.csv; transforms: Saved local samples; matched no-pulse subtraction; explicitly labeled amplitude normalization; no smoothing.

long-baseline [evidentiary]: Do mode energy and charge remain controlled for the full refined baseline?

View: PNG/SVG/PDF and exact CSV; observables: local clock, mode energy, control residuals

usable: matched prediction, parity, scaling and persistent ticks

inadequate: resolved failure or missing ticks

Controls: frozen, numerics; uncertainty: Independent numerical controls; spherical restriction

Data: plot-data/summary.json, plot-data/traces.csv; transforms: Saved local samples; matched no-pulse subtraction; explicitly labeled amplitude normalization; no smoothing.

RESOURCE LIMITS

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{"max_memory_mb": 1024, "max_output_mb": 100, "max_wall_seconds": 1800}
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PROHIBITED INFERENCES

No emergent spacetime, gravity, protected knot or nonspherical stability claim.

Concentric counterpropagating shells are not opposing beams or a recoil calculation.

This ideal local field readout is not a fully invariant two-object Test 7/9 protocol.